

LECTURE ACTIVITY 6: TASTING WATER – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from “Introduction to Statistical Investigations,” by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

- 1) Conduct a simulation-based approach to evaluate claims about a single long-run proportion using the sample proportion as the statistic.
- 2) Apply the correct symbols/words and distinguish between statistic and parameter when referring to a categorical variable
- 3) State null and alternative hypothesis for a statistical test for a long-run proportion in words and symbols (not a 0.5 hypothesized value)
- 4) Calculate, interpret, and evaluate the p -value
- 5) Use the p -value to answer the research question in context.

Key Terms:

- | | | |
|--------------------------|-----------------------|------------------------|
| • Null Hypothesis | • π and $\hat{p}$ | • Strength of Evidence |
| • Alternative Hypothesis | • p -value | |
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Introduction

People spend a lot of money on bottled water. But do they really prefer bottled water to ordinary tap water? Researchers at Longwood University (Lunsford and Dowling Fink, 2010) investigated this question by presenting people in the Longwood community who came to a booth at a local festival with four cups of water. Three cups contained different brands of bottled water (Aquafina, Fiji, Sam’s Choice) and one cup was filled with tap water. Cups were not labeled. Each subject was asked which of the four cups of water they most preferred. Researchers kept track of how many people chose tap water to see whether tap water was chosen less often than would be expected by random chance alone. The researchers found that 3 of 27 subjects selected tap water.

Step 1: Ask a Research Question

The research question we will investigate in this activity is: **Do people in the Longwood community genuinely prefer tap water less than bottled water?**

Step 2: Design a Study and Collect Data

Recall that we have two different viewpoints on data: as samples from a population, or as observations from a random process. In the last lecture activity (“Can Dogs Understand Human Cues?”) we modeled our data as observations from a random process, where one trial of underlying process was “Harley choosing a cup.”

This study could actually be interpreted either way! In particular, there are two ways we could think about an observational unit:

- **Viewpoint 1:** An observational unit is one person
- **Viewpoint 2:** An observational unit is one person picking one of the four cups of water as their favorite



Question 1

- (a) Which viewpoint corresponds to thinking of our data as samples from a population, and which corresponds to thinking of our data as observations from a random process?

Solution: Viewpoint 1 corresponds to “samples from a population” and viewpoint 2 corresponds to “observations from a random process.”

- (b) When thinking of our data as a sample from a population, what is the population of interest? Use the context of the study.

Solution: The population of interest, if we think of one observational unit as one person, is **the set of all people in the Longwood community.**

- (c) What is the sample size under the two different viewpoints? Does the final value of the sample size change depending on which viewpoint you use?

Solution: Whether we think of an observational unit as one person or a trial of “a person picking their favorite of the four cups of water,” the sample size is **27.**

So, thankfully, our value for the sample size does not change depending on which viewpoint we use. This queues us in to realizing that the observed value of the statistic (in this case, a sample proportion) probably won’t change depending on which viewpoint we pick!

Does our notion of the variable in the study change depending on which viewpoint we use?



Question 2

- (a) Under viewpoint 1, identify the variable in the study. Encode your variable as a **binary** variable (remember this term from the reading?), and assign one of the levels to be a “success.”

Solution: Remember that a variable is something that we can measure from each observational unit. Under viewpoint 1, an observational unit is one person - the attribute we're recording from a person is **which cup of water the person chose**. The problem tells us to encode the variable as a binary one, meaning we should use only two levels: **tap water** and **bottled water**. More specifically: we don't (for the purposes of this study) really care which brand of bottled water the person chose, we only care about whether they chose tap water or not. I'll assign **tap water** to be a "success," though you are perfectly allowed to select "bottled water" to be the "success-" just make sure you're consistent.

- (b) Under viewpoint 2, identify the variable in the study. Encode your variable as a **binary** variable, and assign one of the levels to be a "success."

Solution: Again, a variable is something that we can measure from each observational unit. Under viewpoint 1, an observational unit is one trial, and the variable of interest is just the outcome of one trial. In particular, the variable is **which cup of water the subject chose**, whose levels are **tap water** and **bottled water**.

So, at the end of the day, our notion of the variable of interest doesn't really change depending on which viewpoint we use!

Note: We technically could have also rephrased the variable to be "did the person pick tap water or not?" which would have levels "**yes**" and "**no**."

Pro Tip

Going forward, let's set "a person selecting tap water" to be a "success." In the real world, you (typically) get to pick which level you assign to be the "success-" just make sure you're consistent!

Finally, one thing that *does* change depending on which viewpoint we use is that of the **parameter of interest**.

Question 3

- (a) Under viewpoint 1, define the parameter of interest. Make sure you're providing a *verbal* description of the parameter, using the context of the problem.

Solution: When we think of our data as samples from a population, the parameter of interest is a population parameter - in this case, the **proportion of all Longwood citizens that prefer tap water to bottled water**.

- (b) Under viewpoint 2, define the parameter of interest. Make sure you're providing a *verbal* description of the parameter, using the context of the problem.

Solution: When we think of our data as observations from a random process, the parameter of interest is a long-run proportion - in this case, the **long-run proportion that a subject in the study would select tap water**.

Spoiler alert - we'll be able to use either of these parameters to describe our study! For the time being, let's

assign some mathematical symbols to each, just so we can reference them separately if we need.

- Let π_1 denote the parameter of interest under viewpoint 1
- Let π_2 denote the parameter of interest under viewpoint 2

Caution

To stress: π_1 does not necessarily equal π_2 . One is a long-run proportion, and the other is a proportion from a finite population (I won't tell you which one is which, so as not to give away the answer to Question 5 above).

Just like with Harley in the last lecture activity, there are two competing possibilities for why we observed the data that we did:

- **Possibility 1:** nobody in the Longwood community has a real preference among the four water options
- **Possibility 2:** there is an actual preference for bottled water over tap water in the general Longwood community



Question 4

- (a) Under possibility 1 (there is no real preference among Longwood students), what is the value of π_1 ?

Solution: If there is no real preference among Longwood students, we'd expect roughly one-quarter of them to pick the cup with tap water. Why one-quarter? Because subjects were presented with four cups, one of which had tap water. If there is no real preference, we'd expect roughly equal proportions of students to pick each of the four options, only one of which is tap water. Hence, under possibility 1, $\pi_1 = 0.25$.

- (b) Under possibility 2 (there is a general preference against tap water in the general Longwood community), what is the range of possible values for π_1 ?

Solution: If there is a general preference against tap water, we'd expect fewer than one-quarter of students to report preferring the cup with tap water to the other cups (with bottled water). Hence, under possibility 1, $\pi_1 < 0.25$.



Question 5

- (a) Under possibility 1 (there is no real preference among Longwood students), what is the value of π_2 ?

Solution: Under the viewpoint of our data as observations from a random process, possibility 1 corresponds to our "chance model-" when there is no real preference in the Longwood community, then each person picks one of the four cups completely randomly. This means that the probability a particular person picks the cup with tap water is one-fourth: $\pi_2 = 0.25$.

- (b) Under possibility 2 (there is a general preference against tap water in the general Longwood community), what is the range of possible values for π_2 ?

Solution: If there is a general preference against tap water, we'd expect the long-run proportion of subjects who select the cup with tap water to be smaller than what we'd expect by chance alone: $\pi_2 < 0.25$.

🔑 Key Takeaway

Regardless of whether we use π_1 or π_2 as our parameter of interest, their values and possible range of values under the various possibilities (possibility 1 and 2) are the same. This is what I meant above when I said “we’ll be able to use either of these parameters to describe our study!”

However: this is also why I’m such a stickler about **verbally defining your parameters**. Don’t just write π and expect your reader to know what you’re talking about - provide a verbal definition of what π means. I will deduct points on quizzes and exams for failing to do this!

Concept Overview

Now that you’ve read Section 1.2 from the textbook, you know that our two “possibilities” are really our null and alternative hypotheses.

- The **null hypothesis** typically represents the “by-chance-alone” explanation. The chance model (or “null model”) is chosen to reflect this hypothesis.
 - We use the symbol H_0 to represent the null hypothesis
- The **alternative hypothesis** typically represents the “there is an effect” explanation that contradicts the null hypothesis.
 - We use the symbol H_a to represent the alternative hypothesis

In the context of this study,

- **Null Hypothesis:**
 H_0 : the long-run proportion of a subject choosing tap water as their favorite is 0.25 (1/4).
- **Alternative Hypothesis:**
 H_a : the long-run proportion of a subject choosing tap water as their favorite is less than 0.25 (1/4).

Step 3: Explore the Data



Question 6

Report the observed value of the statistic as a proportion. Is this in the direction of the alternative hypothesis?

Solution: The observed value of the statistic is the sample proportion of the 27 students who picked tap water: $\hat{p} = 3/27 = 0.1111$

Concept Overview

As we've done above, we often use mathematical symbols to represent quantities and simplify our writing. Throughout this course we'll emphasize written explanations, but will also show you mathematical symbols which you are free to use as a short-hand once you are comfortable with the material.

The distinction between a **parameter** and **statistic** is so important that we always use different symbols to refer to them.

- When dealing with a **parameter** that is either a long-run proportion or a population proportion, we use the Greek letter π (pronounced “pie,” or “pee”)
- When working with a **statistic** that is the sample proportion, we use the symbol $\hat{p}$ (pronounced “p-hat”)
- When a specific numerical value for the **sample size** is not provided, we use n as a placeholder

So, in the context of this study:

- We'll set the parameter to be π_{tap} , the (unknown) long-run proportion that a subject will choose tap water as their favorite (or, if we're viewing our data as samples from a population, the proportion of all Longwood students that prefer tap water to bottled water).
 - Note: we used the subscript “tap” to remind ourselves of what we are considering as “success” in the levels of the variable.
- The statistic is $\hat{p}_{\text{tap}}$, the sample proportion of the 27 subjects who chose tap water. Its numerical value, given the data, is $\hat{p}_{\text{tap}} = 0.111$.
- The sample size is $n = 27$, as we identified in Question 1 above.



Question 7

Hypotheses are always conjectures about the unknown parameter π_{tap} . Restate the null and alternative hypotheses using the appropriate symbols. **Hint:** you should never be writing $H_0 = \dots$ because a hypothesis cannot be “equal” to anything - it is a conjecture about the unknown parameter.

Solution: The null hypothesis is $H_0: \pi_{\text{tap}} = 0.25$ and the alternative hypothesis is $H_A: \pi_{\text{tap}} < 0.25$, where π_{tap} is the (unknown) long-run proportion that a subject will choose tap water as their favorite (or, if we're viewing our data as samples from a population, the proportion of all Longwood students that prefer tap water to bottled water).

⚠ Caution

Be careful with your notation! Firstly, don't forget to define your parameter verbally somewhere in your work. Secondly, the null and alternative hypotheses should be **statements about parameters or quantities of interest**. Don't make the mistake of writing things like $H_0 = 0.25$ as this is incorrect notation, and may result in point deductions on quizzes and exams.

Step 4: Draw Inferences Beyond the Data

We will now use simulation to investigate how surprising the observed sample proportion of 0.111 (3 of 27 selecting tap water) would be if in fact subjects did not dislike tap water and so each had a $1/4$ probability of selecting tap water. (Note also that our null model assumes the same probability for all subjects.)

For this study, we cannot use a simple coin toss to simulate the subjects' choices, as we did with the study in the previous lecture activity ("Can Dog Understand Human Cues?") because the null hypothesis probability is not $1/2$ anymore. Instead, we need a different (easily repeatable) random process to model the random selection. There are a couple of options:

- **Spinners:** use a spinner with two regions, black and red. Let the black region represent a "success" (in the context of the study, picking tap water) and the red region represent "failure" (in the context of the study, picking bottled water)
- **Poker Chips:** use red and black poker chips, with black poker chips representing "successes" (choosing tap water) and red poker chips representing "failures" (choosing bottled water)



Figure 1: A spinner with red and black regions.

Our textbook uses spinners, so let's stick with that analogy. Because the value of the parameter is 0.25 under the null, we should set up our spinner so that exactly one-quarter of its area is black and three-quarters of its area is red (see the figure to the left). This ensures that, with each spin, there is a one-fourth chance of observing a success (landing on black). Remember: we want each spin to mimic one trial from the underlying process, under the null distribution. This will enable us to simulate more sample proportions under the null without having to actually collect more real-world data.

Pro Tip

The phrase "under the null" is just shorthand for saying "assuming the null hypothesis is true." As we will see over the next few lectures, much of statistical inference is conducted under the null.



Question 8

How many spins of the spinner will we need in order to simulate one repetition of the experiment when there is equal preference among the four types of water, under the null?

Solution: We want the number of spins in each repetition to match the sample size: **27**.

We can use the [One Proportion Applet](https://www.rossmanchance.com/applets/2021/oneprop/OneProp.htm) (for those of you with a printed copy of this handout, here is the url: <https://www.rossmanchance.com/applets/2021/oneprop/OneProp.htm>) to carry out this simulation.



Question 9

- (a) Use the One Proportion Applet to carry out one repetition of the chance model, using the following steps:

1) Under "Probability of heads/probability of success," enter the value 0.25 (again, make

sure you understand why). Note that the applet will stop using the language of coin tossing.

- 2) Enter the appropriate “sample size” (number of subjects/trials in this study).
- 3) Enter 1 for the number of samples, and press “Draw Samples.”

Do not close out of the applet! We’ll make changes to the configurations in the next few parts/questions.

- The number of successes I observed is 7 **Answers may vary**
- (b) Navigate back to the applet, and under the “Choose Statistic” menu click the radio button that says “Proportion of successes.” The dotplot should now display the sample proportion, as opposed to the sample count. Report the proportion of successes in this simulated sample, and verify how it was computed (that is, calculate it with a calculator and check that it matches the value that appears on the dotplot in the applet).

- The proportion of successes I observed is 0.2593 **Answers may vary**

- (c) Leaving the “Proportion of Successes” ratio button selected but unchecking the “Animate” box, click on “Draw samples” four more times (to complete a total of five repetitions). Do you get the same results each time? Why not? **Still do not close out of the applet!**

Solution: No, I do not get the same proportions each time, because of **sampling variability**.

- (d) Now enter 9995 for the “Number of samples” and check the “hide spinners” box. Click on “Draw Samples,” bringing the number of simulated samples to 10,000. Take a screenshot/sketch the plot with 10,000 different simulated samples and paste it in the space below. **Still do not close out of the applet!**

Solution: Answers may vary - here’s my plot:

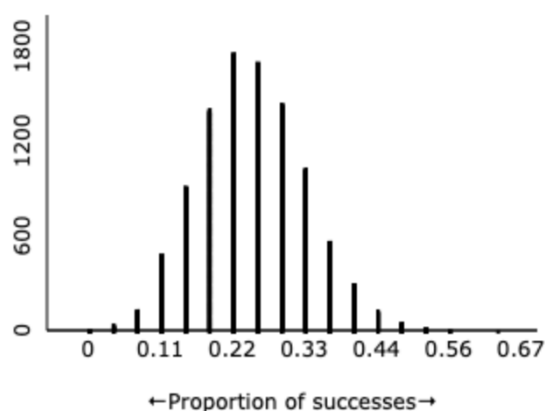


Figure 2: Example dotplot from the applet, resulting from 10,000 simulated samples of size 27.

Concept Overview

The distribution of simulated sample proportion is called the **null distribution**, because it is based on assuming the null hypothesis to be true. We can see that the null distribution is bell-shaped and centered around 0.25 (the **null value**; i.e. the value of the parameter under the null hypothesis).



Question 10

Recall that the observed statistic (proportion who selected tap water in the sample) was $\hat{p}_{\text{tap}} = 3/27 \approx 0.1111$. Is this a typical value for the null distribution you have simulated? Or is this value far in the tail of the null distribution?

Solution: The observed statistic, 0.111, is somewhat in the left tail. It is lower than typical, but not extremely far out.

Concept Overview

You might very well find that Question 10 is a bit of a tough call. The value 0.111 is not far in the tail of the distribution, but it is also not near the middle of the distribution. To help make a judgment about strength of evidence in this case, we can count how many (and what proportion) of the simulated sample proportions are as extreme or more extreme than the observed statistic. This proportion is an approximation to what is referred to as a **p-value**.



Pro Tip

In a future lecture, we'll see how a true *p*-value is actually a probability, hence why I described our proportion above as an *approximation* to the *p*-value.



Question 11

Return to the plot you created in Question 9 above. We'll now use the applet to calculate an approximate *p*-value.

- 1) Click the " $\geq$ " sign to change it to " $\leq$ " (to match the alternative hypothesis)
- 2) Enter the observed statistic (sample proportion) who chose tap water in the box to the left of the "Count" button
- 3) Click on the "Count" button, which should now enable you to see the estimate of the *p*-value as a proportion in red immediately below the "Proportion of samples."

Record the number and proportion of simulated sample proportions that are as extreme or more extreme than the observed statistic. **To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) estimated *p*-value.**

- My observed **number** of samples with sample proportions as or more extreme as the observed statistic is 666
- My observed **proportion** of samples with sample proportions as or more extreme as the observed statistic is 0.0666

Here is my screenshot:

Solution: Answers may vary - here's my screenshot:

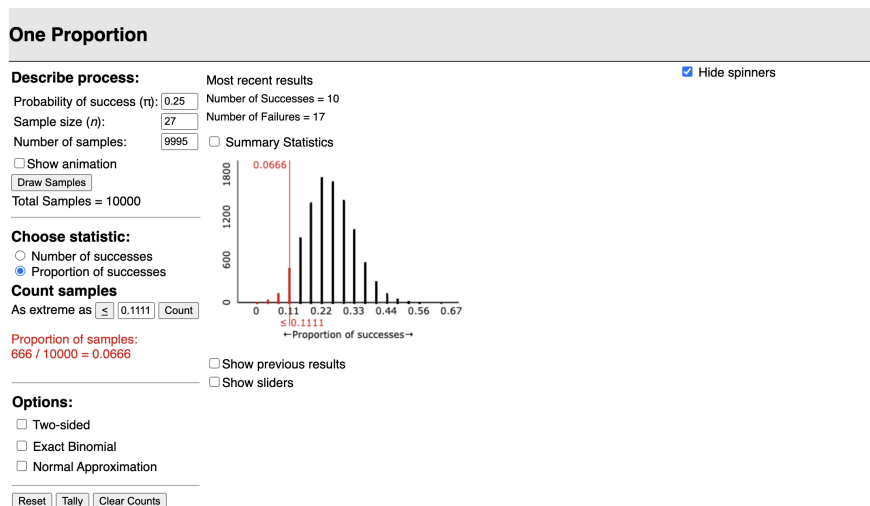


Figure 3: Example screenshot from the applet, indicating my approximate p -value.

Alright, so now we have an approximate p -value. But what do we *do* with it? In particular, it's not enough to just get (or approximate) a p -value - we need to **evaluate** the value and **interpret** it as well.

Let's tackle the evaluation step first. How do we evaluate this p -value as a judgment about strength of evidence provided by the sample data against the null hypothesis? One answer is: The smaller the p -value, the stronger the evidence against the null hypothesis and in favor of the alternative hypothesis. But how small is small enough to regard as convincing evidence? **There is no definitive answer**, but here are some guidelines:

p -value	Strength of Evidence
$0.10 < p$ -value	Not much evidence against the null hypothesis; null is plausible
$0.05 < p$ -value ≤ 0.10	Moderate evidence against the null hypothesis
$0.01 < p$ -value ≤ 0.05	Strong evidence against the null hypothesis
p -value ≤ 0.01	Very strong evidence against the null hypothesis

Table 1: Guidelines for evaluating the strength of evidence from a p -value

⚠ Caution

This is only a set of **suggested guidelines**. In statistics, we try to avoid relying on cutoffs as much as possible because they can be so arbitrary. I encourage you to think more holistically about p -values, as a “sliding scale of strength-” the smaller the p -value, the stronger the evidence against the null.

On the interpretation side, the approximate p -value is the proportion of the simulated sample proportions that are equal to the observed statistic or more extreme, under the null. The “more extreme” values are

always measured in the direction of the alternative hypothesis. That is why we considered all the simulated values equal to 0.1111 or smaller.

Question 12

Using your approximate p -value from Question 11 above and the guidelines shown above, how much evidence do the data provide against the null hypothesis? Choose one.

- ☐ Not much evidence against the null hypothesis
- ☒ **Moderate evidence against the null hypothesis**
- ☐ Strong evidence against the null hypothesis
- ☐ Very strong evidence against the null hypothesis

Solution: My approximate p -value was 0.0666, which falls under the “moderate evidence against the null hypothesis” category. Your answers may vary depending on your approximate p -value.

Step 5: Formulate Conclusions

Alright, we’re finally in a position to formulate a conclusion to our initial research question: do people in the Longwood community genuinely prefer tap water less than bottled water?

Question 13

Answer the research question based on the analysis you conducted. Some things to keep in mind:

- Don’t just say “yes” or “no;” reference the results of Step 4 (Drawing Inferences) as much as possible.
- Refrain from using the word “prove-” remember that hypothesis testing just gives us a sense of which hypothesis is *more plausible*. We are never able to “prove” or “disprove” either the null or alternative hypotheses.

Solution: The data do NOT provide strong evidence that people in the Longwood community genuinely prefer tap water less than bottled water. The results from the study (sample proportion of 0.111 subject choosing tap water in the sample) was NOT that unlikely in the distribution that assumes people are choosing the cups at random. It MIGHT be that the results of this study could have happened by chance.

Pro Tip

We’ll continue to refine our conclusions as we progress through the next few sections/chapters of the textbook.

Question 14

Consider the following paragraph, taken from the article:

“Our data were collected at Oktoberfest, which is an event attended primarily by students involved in extracurricular organizations (such as fraternities, sororities, and clubs), some alumni, a few faculty and staff advisors, and the occasional child.”

(Lunsford and Dowling Fink, 2010)

- (a) What is the sampling frame in the context of this study?

Solution: The sampling frame is the set of all attendees of the Longwood Oktoberfest.

- (b) Remember that Step 5 is the “scope of inference” step of the investigative process. As such, it’s important to critically consider the extent to which we’re able to generalize our findings. Do you think we can generalize the findings of our study to the entire Longwood population? If not, to what population do you think we can generalize our findings?

Solution: It might be a little risky to generalize our findings to the general Longwood community; it’s safer to generalize our findings only to the set of attendees at the Longwood Oktoberfest. For example, what if attendees of Oktoberfest have heightened (or less heightened) palattes than members of the broader Longwood community? In this case, their ability to detect the tap water from the bottled water may not be representative of the abilities of the members of the broader Longwood community.

Step 6: Look Back and Ahead



Question 15

A single study will not provide all of the information needed to fully understand a broad, complex research question. Thinking back to the original research question, what additional studies would you suggest conducting next?

Solution: I would suggest repeating the study with a larger sample, testing people in different locations, using more water brands, randomizing cup order, and checking whether people can identify tap water versus bottled water.